Other Adverse Effects

Hypertrichosis occurs in virtually all patients on long-term cyclosporine (CsA) therapy. It is not limited to androgen-dependent, hair-bearing areas and does not show a tendency toward spontaneous remission.

Gingival hypertrophy is reported in up to 70% of CsA-treated patients and is more common in children, individuals with poor oral hygiene, and those concomitantly using calcium channel blockers. Improvement in gingival overgrowth has been observed with topical or systemic azithromycin and metronidazole.

An **acneiform eruption**, clinically indistinguishable from steroid-induced acne, is frequently reported. A disseminated comedonal or cystic acneiform eruption may also occur. These dermatologic side effects can appear at any time during CsA therapy, though they are most commonly observed at treatment initiation.

Other cutaneous manifestations include **keratosis pilaris**, **sebaceous hyperplasia**, **warts**, and **epidermal inclusion cysts**, which occur in up to one-third of CsA-treated patients.

Osteoporosis may result from CsA's effects on osteoblasts and osteoclasts, as well as alterations in lymphokine release.

CsA can induce **hyperuricemia** in up to 15% of patients, which may also serve as an early indicator of CsA-induced nephrotoxicity.

Myopathy has been reported in transplant patients receiving high doses of CsA. Caution is advised when combining CsA with statins due to an increased risk of myotoxicity.

Pregnancy

CsA does not appear to be mutagenic or teratogenic. However, in pregnant women taking CsA—particularly transplant recipients—there is an increased incidence of **preterm birth**, **fetal growth retardation**, **abortion**,

preeclampsia, and **hypertension**. These risks are heightened in transplant patients.

No neonatal complications have been reported in children born to fathers receiving CsA.

CsA crosses the placenta and is excreted in breast milk. Therefore, **adequate contraception** is recommended for women of childbearing potential.

CsA-treated transplant patients have a lower relative risk of life-threatening infectious complications compared to those receiving azathioprine and prednisolone. Nonetheless, **increased vigilance for infections** is warranted in all CsA-treated patients.

Tacrolimus (TCL)

Tacrolimus (TCL), a calcineurin inhibitor formerly known as FK506, is a macrolactam drug initially isolated from *Streptomyces tsukubaensis*.

Mechanism of Action

The proposed mechanisms of action of TCL are similar, if not identical, to those of CsA (see Macrolactam Drugs).

Pharmacokinetics

TCL can be administered **orally**, **topically**, and **intravenously**. Systemic pharmacokinetics follow a two-compartment model, characterized by a rapid initial distribution phase and a terminal elimination half-life of 12 to 21 hours. TCL is primarily metabolized in the liver, with less than 1% excreted unchanged in urine.

While TCL has not been shown to be more effective than CsA, it demonstrates **better bioavailability**. Topical formulations are available as 0.3% and 0.1% ointments; for additional details, refer to Chapter 221.

Dosing Regimen

	ORAL	INTRAVENOUS
Adults	150–200 µg/kg/day	25–50 µg/kg/day
Children	200–300 µg/kg/day	50–100 µg/kg/day

Initiating Therapy

In patients with hepatic dysfunction, the systemic TCL dose should be reduced by 10% to 30% from the standard regimen.

Monitoring Therapy

Whole blood concentrations of TCL can be reliably monitored using either **enzyme-linked immunosorbent assay (ELISA)** or **HPLC mass spectrometry**, with good correlation between the two methods.